

UNIT – 3	9 Hours
Analysis of AC circuits	
Instantaneous value, RMS values, Average value, Form factor, Peak factor (sine and square waveform).	
Analysis of R, L, C, RL, RC and RLC series circuits.	

Analysis of AC Circuits:

Alternating current (AC) is an electric current that periodically reverses direction and changes its magnitude continuously with time.

Some of the examples of AC waveform are shown in Figure 3.1

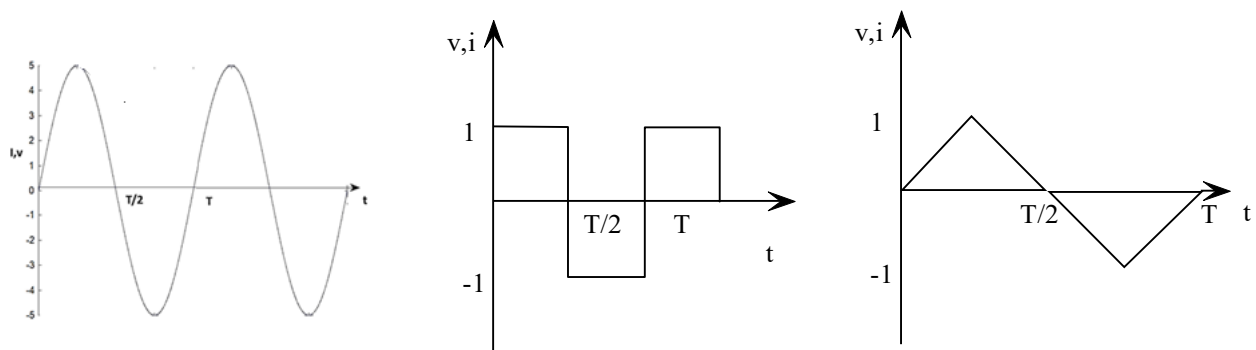


Fig 3.1 Examples of AC waveforms

Terminologies used in AC sine wave representation: ωt (θ)

Consider the sine wave shown in Figure 3.2

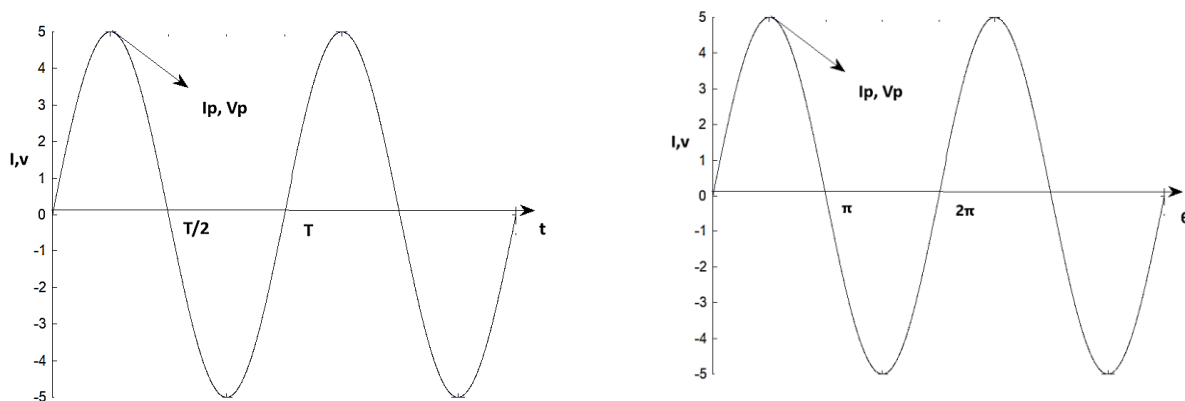


Fig 3.2 Sinusoidal voltage or current waveform

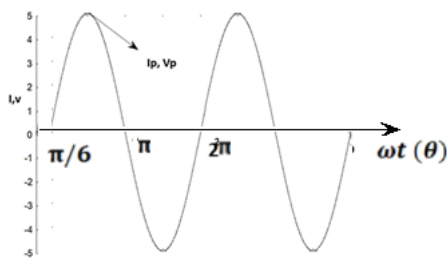
- (i) **Cycle:** When an Alternating quantity goes through complete set of positive and negative values or goes through 360 electrical degrees.
- (ii) **Time Period (T) :** Time taken to complete one cycle by AC wave. It is represented by T (sec)

(iii) **Frequency (f)** : Number of cycles made per second. It is the reciprocal of Time period ($\frac{1}{T}$) and its unit is Hz (Hertz)

(iv) **Amplitude:** Maximum value attained by an alternating quantity in one cycle and also called Peak Value / Max. Value.

(v) **Phase:**

If the sine wave is shifted by $\pi/6$ from the zero point of reference as shown in Figure 3.3, then phase is said to be $\pi/6$ radians



Instantaneous value: The magnitude of a waveform at any instant of time with reference to zero. Instantaneous values are denoted by lower case symbols such as e, v and i etc.

Consider the current waveform shown in Fig 3.3

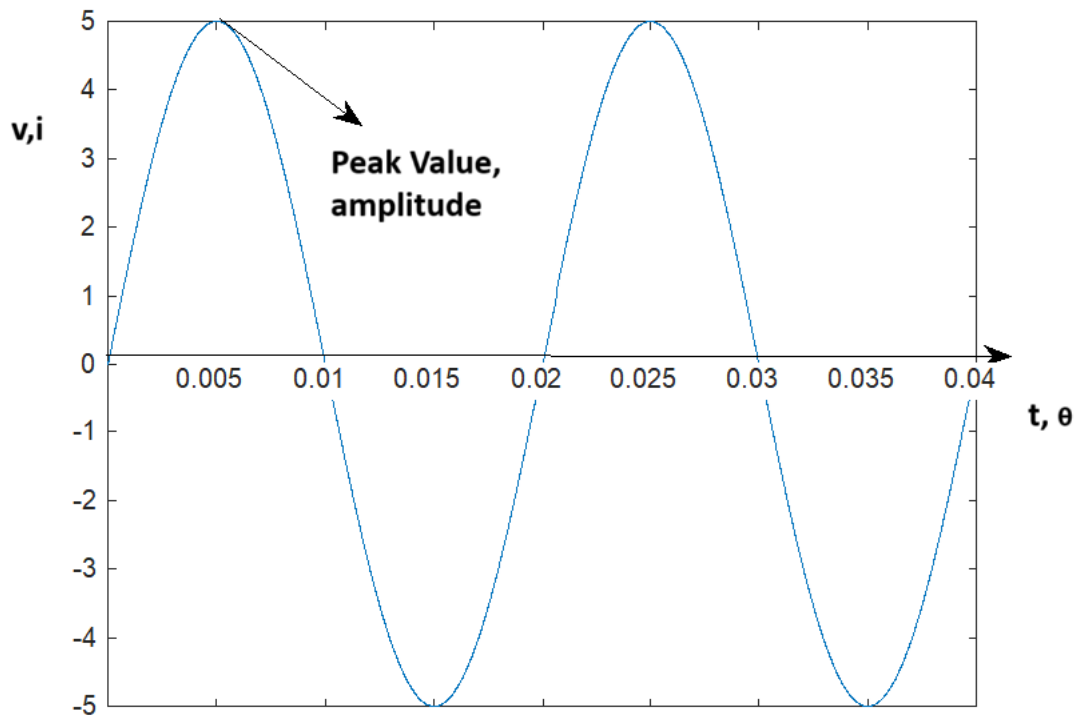


Fig 3.3 Sinusoidal voltage or current wave

The waveform is defined as $i = 5 * \sin(\omega t) = 5 * \sin (314t)$ or $v = 5 * \sin (314t)$

where $\omega = 2 * \pi * f = 2 * \pi * \frac{1}{T}$ and its unit is rad/sec

5 is the amplitude of the wave.

Instantaneous current at

$t=0$, $i=0$

$t=0.005$ S, $i=5$ A

$t=0.01$ S, $i=0$ A

$t=0.015$ S, $i=-5$ A

RMS value:

"RMS" stands for "Root-Mean-Squared", also called the effective or heating value of alternating current.

If $i = I_m \sin \theta$ is passed through a resistor having resistance R ohms, the heating effect of 'i' is $i^2 R$.

Thus, the RMS value of an alternating current is defined as that steady (D.C) current which when flowing through a given resistor for a given time produces the same amount of heat as produced by the alternating current when flowing through the same resistor for the same period of time.

In other words, the R.M.S value is defined as the square root of means of squares of instantaneous values.

Consider the current waveform shown in Figure 3.4

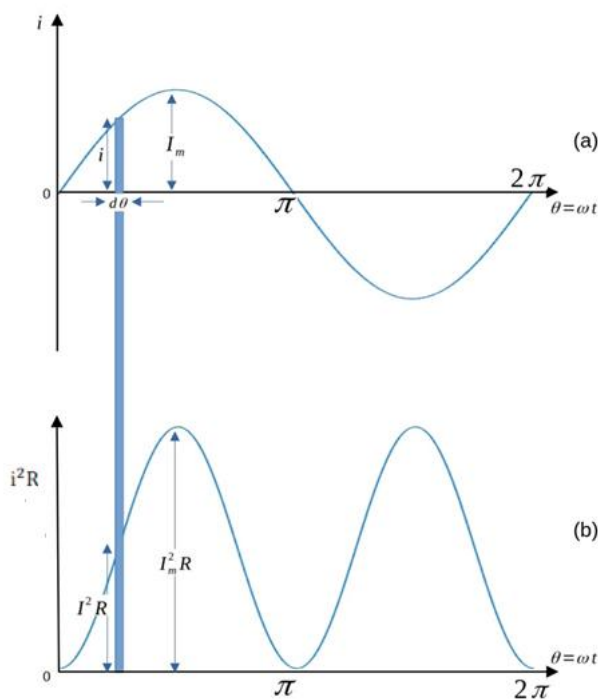


Fig 3.4 i and $i^2 R$ wave

If the current is passed through a resistor having resistance R ohms, instantaneous heating effect is given by $i^2 R$ watts.

Average heating effect over half-cycle is

$$\frac{\text{Area enclosed by } i^2 R \text{ over half cycle}}{\text{Length of base}}$$

Consider a small elementary strip of length $d\theta$ in $i^2 R$ wave as shown in Figure 3.4 (b). During interval $d\theta$ radians, heat generated is $i^2 R d\theta$ watt radians and is represented by the area of the shaded strip. Hence heat generated during the first half-cycle is the area enclosed by the $i^2 R$ curve and is given by

$$\begin{aligned} \int_0^\pi i^2 R d\theta &= I_m^2 R \int_0^\pi \sin^2 \theta d\theta = I_m^2 R \int_0^\pi \left(\frac{1 - \cos 2\theta}{2} \right) d\theta \\ &= I_m^2 R \left[\frac{\theta}{2} - \frac{\sin 2\theta}{4} \right]_0^\pi = \pi R \frac{I_m^2}{2} \\ &= \end{aligned}$$

Then Average Heating effect is
$$\frac{\pi R \frac{I_m^2}{2}}{\pi} = R \frac{I_m^2}{2}$$

If 'I' is the value of direct current through the same resistance to produce the same heating effect as that of ac

$$I^2 R = R \frac{I_m^2}{2}$$

Or

$$I = \frac{I_m}{\sqrt{2}} = 0.707 I_m$$

Note: The above derivation is valid for Voltage also. $V = \frac{V_m}{\sqrt{2}}$

Average Value:

This is defined on the basis of amount of charge transferred, which is given by $q = I t$.

The average value of an alternating current is equal to that (D.C) steady current, which transfers the same amount of charge as transferred by the alternating current across the same circuit and in the same time.

Or, alternatively, average value of current is

$$\frac{\text{Area enclosed over a cycle}}{\text{Length of base over a cycle}}$$

Average value of a sinusoidal wave: The average value of a sinusoidal wave is always zero.

Average value of a fully rectified sinusoidal wave:

Consider the rectified sinusoidal wave with an elemental rectangular strip of length $d\theta$ as shown in Figure 3.5

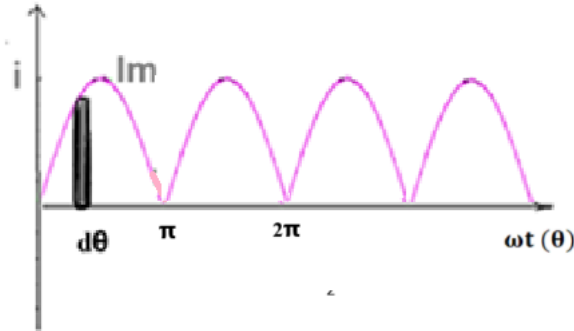


Fig 3.5 Fully Rectified sine wave

For a very small interval $d\theta$ radians, the area of the shaded strip is $i.d\theta$ ampere radians. Therefore, total area enclosed by the current wave over half-cycle is

$$\int_0^{\pi} i d\theta = \int_0^{\pi} I_m \sin\theta d\theta = I_m [-\cos\theta]_0^{\pi} = 2I_m \text{ ampere radians}$$

Thus, average value of current (I_{av}) is

$$\frac{\text{Area enclosed over a cycle}}{\text{Length of base over a cycle}} = \frac{2I_m}{\pi} = 0.637 I_m$$

Thus, the average value of sinusoidal quantity is 0.637 times its maximum value.

Note: Similarly, the average voltage is given by $V_{av} = 0.637 V_m$

Average value of a half rectified sinusoidal wave:

Consider the rectified sinusoidal wave with an elemental rectangular strip of length $d\theta$ as shown in Figure 3.6

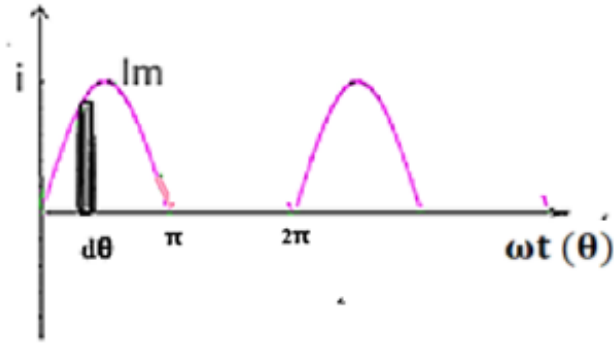


Fig 3.6 Half Rectified sine wave

For a very small interval $d\theta$ radians, the area of the shaded strip is $i \cdot d\theta$ ampere radians. Therefore, total area enclosed by the current wave over half-cycle is

$$\int_0^{\pi} i \, d\theta = \int_0^{\pi} I_m \sin \theta \, d\theta = I_m [-\cos \theta]_0^{\pi} = 2I_m \text{ ampere radians}$$

Thus, average value of current is

$$\frac{\text{Area enclosed over a cycle}}{\text{Length of base over a cycle}} = \frac{2I_m}{2\pi} = 0.318 I_m$$

Average value of a square waveforms:

(i) Consider the square waveform shown in Fig 3.7

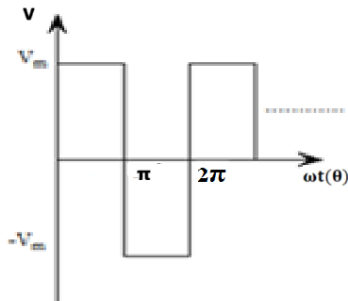


Fig 3.7 Square Wave

The average value is the mathematical mean of all the waveforms points over the period of one cycle. Hence average value over a complete cycle is zero.

(ii) Consider the square waveform shown in Fig 3.8

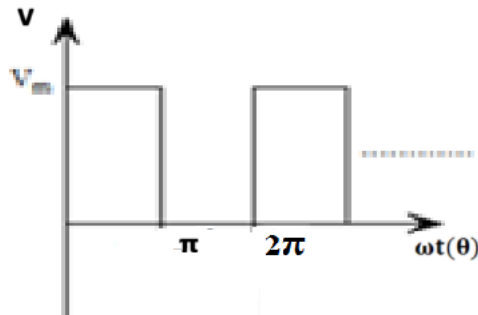


Fig 3.8 Square Wave

Thus, average value of Voltage is

$$\frac{\text{Area enclosed over a cycle}}{\text{Length of base over a cycle}} = \frac{V_m \pi}{2\pi} = 0.5 V_m$$

Form Factor:

The ratio of RMS value to average value of an alternating quantity is called form factor. For fully

rectified sine wave current, $K_f = \frac{I}{I_{\text{avg}}} = \frac{I_m / \sqrt{2}}{2I_m / \pi} = 1.11$

Peak Factor:

The ratio of maximum value to the RMS value of an alternating quantity is called peak factor For sine wave current,

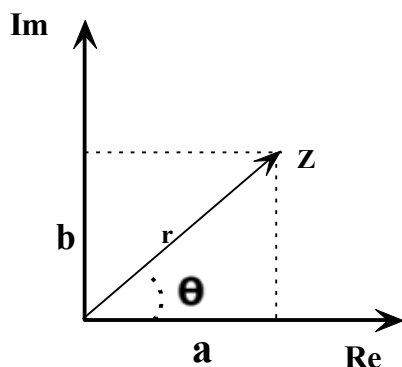
$$K_p = \frac{I_m}{I_{\text{rms}}} = \frac{I_m}{I_m / \sqrt{2}} = 1.414$$

Complex Number Representation:

A complex quantity can be represented in three forms. They are

- (i) **Rectangular form (Cartesian Form):** A complex number z is written as $z = a + bi$, where a and b are real numbers. 'a' represents the real part ($\text{Re}(z)$) and 'b' represents the imaginary part ($\text{Im}(z)$).

Consider the vector shown in Figure.....



$z=a+jb$ is the rectangular form representation.

- (ii) **Polar Form:** A complex number z can also be represented using its magnitude (or modulus) r and angle (or argument) θ .

$z = r\angle\theta$ is the polar form representation where $r = \sqrt{a^2 + b^2}$ and θ is calculated as $\theta = \tan^{-1}(b/a)$.

- (iii) **Exponential Form:** Based on Euler's formula $e^{j\theta} = \cos(\theta) + j\sin(\theta)$, a complex number can be written in exponential form as $z = re^{j\theta}$. This form combines the modulus r and the angle θ .

Exercise : Convert the following complex quantities into other forms

(i) $z=3+4j$ $z = 5\angle 53.13^\circ$ $z = 5e^{j53.13}$

(ii) $z=3-4j$ $z = 5\angle -53.13^\circ$ $z = 5e^{-j53.13}$

(iii) $z=-3-4j$ $z = -5\angle 53.13^\circ$ $z = -5e^{j53.13}$

(iv) $z=-3+4j$ $r = \sqrt{3^2 + 4^2} = 5$

$$\theta = \tan^{-1}\left(\frac{4}{-3}\right) = 180 - \tan^{-1}\left(-\frac{4}{3}\right) = 180 - 53.13 = 126.87$$

$$z = 5\angle 126.87^\circ \quad z = 5e^{j126.87}$$

Power and Power factor:

Power in AC circuits is defined as the amount of energy delivered per unit of time by the circuit. A circuit element dissipates or produces power according to $P=IV$, where I is the current through the element and V is the voltage across it. Since the current and the voltage both depend on time in an ac circuit, the instantaneous power $p(t)=i(t)v(t)$ is also time dependent.

There are three classification of ac power:

- 1. Active power (Real Power):** This is the power that performs useful work, such as running machinery, lighting, or heating equipment. It is given by $P=VI \cos(\phi)$ and its unit is Watt (W), KW or MW.
- 2. Reactive Power:** Reactive power is the power that comes back and forth between the source and the load. It does not contribute to useful work. It is given by $Q=VI \sin(\phi)$ and its unit is VAR, KVAR or MVAR.
- 3. Apparent Power:** This is the total power drawn by the system, a combination of real and reactive power. It is given by $S=VI=\sqrt{P^2 + Q^2}$ and its unit is VA, KVA or MVA

The power triangle is shown in Figure 3.9

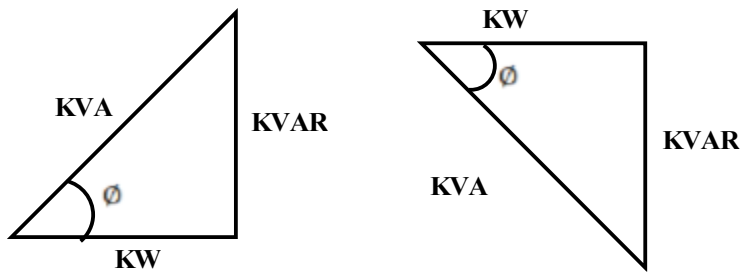


Fig 3.9 Power Triangle

Power factor: The ratio of the active power to an apparent power existing in the circuit is known as the power factor of the alternating current. Power factor (PF) is a dimensionless quantity, ranging from 0 to 1, that measures how effectively an electrical system uses the power it receives. It is given by $\cos(\phi)$ where ϕ is the angle by which current lags the voltage or advances the voltage (popularly known as phase angle of the circuit)

Analysis of a purely resistive circuit:

Consider a circuit having resistance R ohms connected across the terminals of an a.c. supply, as in Figure 3.10. In the circuit shown R is a pure resistance to which an alternating voltage

$v = V_m \sin \omega t$ is applied, due to which an alternating current i flows through it.

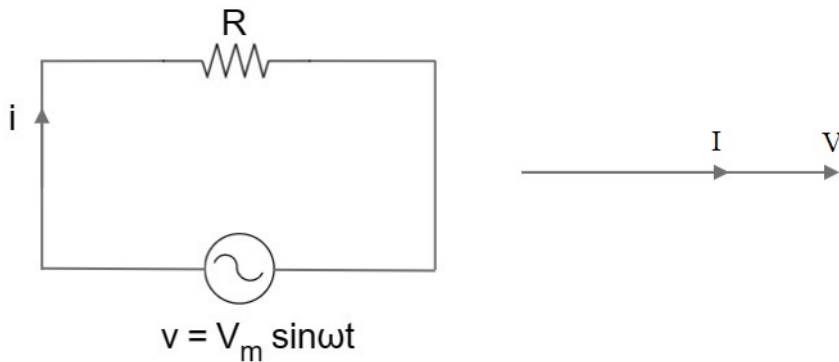


Fig 3.10 AC sinusoidal voltage applied to Resistance R

By Ohms law,

$$i = \frac{v}{R} = \frac{V_m \sin(\omega t)}{R} = I_m \sin(\omega t) \text{ where, } I_m = \frac{V_m}{R}$$

By observing the equations for voltage, $v = V_m \sin \omega t$ and $i = I_m \sin \omega t$, it can be concluded that the current is in phase with the voltage. Vectorially the RMS values of voltage and current are represented as shown in Figure 3.9

The instantaneous power consumed by the resistance is given by,

$$p = v \times i = V_m \sin \omega t I_m \sin \omega t = V_m I_m \sin^2 \omega t$$

$$p = V_m I_m \left(\frac{1 - \cos 2\omega t}{2} \right) = \frac{1}{2} V_m I_m - \frac{1}{2} V_m I_m \cos 2\omega t$$

Equation consists of two parts. The second part $\frac{1}{2} V_m I_m \cos 2\omega t$ is a periodically varying quantity whose frequency is two times the frequency of the applied voltage and its average value over a period of time is zero. Power is a scalar quantity and hence only its average value has to be taken into account. Hence the power consumed by the resistance is only due to the first part $\frac{1}{2} V_m I_m$.

∴ Average Power,

$$P = \frac{1}{2} V_m I_m = \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} = VI$$

The waveforms of v , i and p are as shown in the Figure 3.11

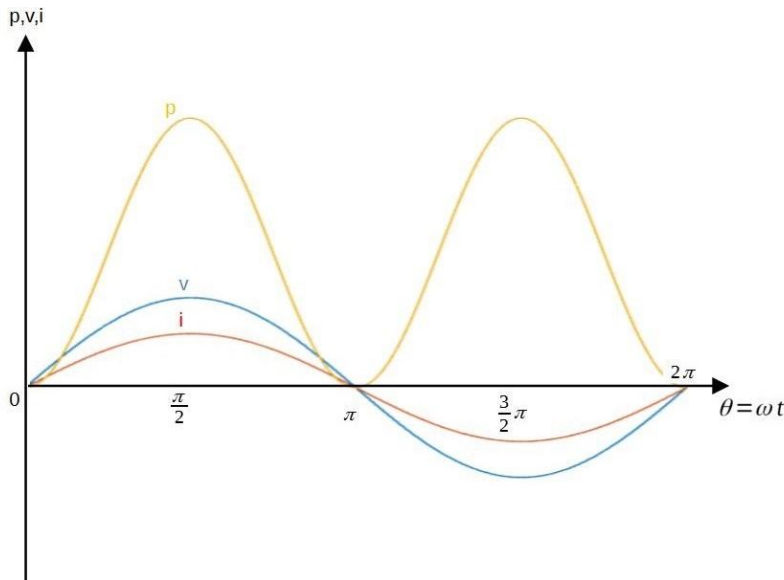


Fig 3.11 Waveforms of voltage, current and instantaneous power in pure resistive circuit

Note: Current is in phase with voltage. ϕ is zero. Hence power factor is unity.

Analysis of a purely inductive circuit:

Consider a circuit with a coil of pure inductance L henry connected across the terminals of an a.c. supply, as in Figure 3.12.

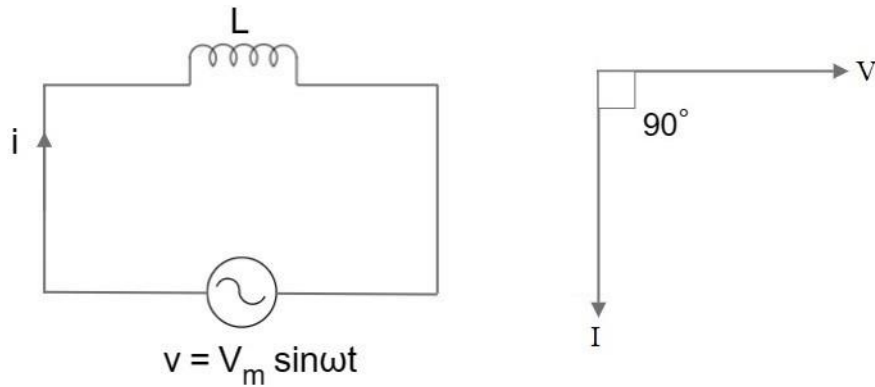


Fig 3.12 a) Pure inductive circuit b) Phasor Diagram

Due to the applied alternating voltage $v = V_m \sin(\omega t)$, an alternating current i flows through it. This alternating current produces an alternating flux, which links the coil and hence an e.m.f e is induced in it, which opposes the applied voltage and is given by,

$$e = -L \frac{di}{dt} = -v$$

$$\therefore v = L \frac{di}{dt}$$

$$di = \frac{v}{L} dt = \frac{1}{L} V_m \sin(\omega t) dt$$

Integrating the above equation, we get

$$\begin{aligned} i &= \frac{V_m}{L} \int \sin(\omega t) dt \\ &= \frac{V_m}{\omega L} (-\cos(\omega t)) \\ &= \frac{V_m}{\omega L} \sin(\omega t - \frac{\pi}{2}) \\ i &= I_m \sin(\omega t - \frac{\pi}{2}) \end{aligned}$$

where
$$I_m = \frac{V_m}{\omega L} = \frac{V_m}{X_L}$$

$$X_L = \omega L = 2\pi fL = \text{reactive inductance in Ohms}$$

By observing the equations for voltage and current, we find that the current lags the voltage by an angle $\pi/2$. Vectorial representation of the RMS values of voltage and current is as shown in Figure 3.12 (b)

The instantaneous power is given by,

$$p = v \cdot i = V_m \sin \omega t \cdot I_m \sin(\omega t - \frac{\pi}{2})$$

$$p = V_m I_m \sin \omega t (-\cos \omega t) = -\frac{1}{2} V_m I_m \sin 2\omega t$$

The equation for p consists of a quantity which is periodically varying and having a frequency two times the frequency of the applied voltage and whose average value is zero. The power consumed by a pure inductor is zero, because the power is a scalar quantity and only its average value has to be considered. The power given to the pure inductance is stored in the form of an electromagnetic field. The waveforms of instantaneous voltage, current & power are as shown in Figure 3.13.

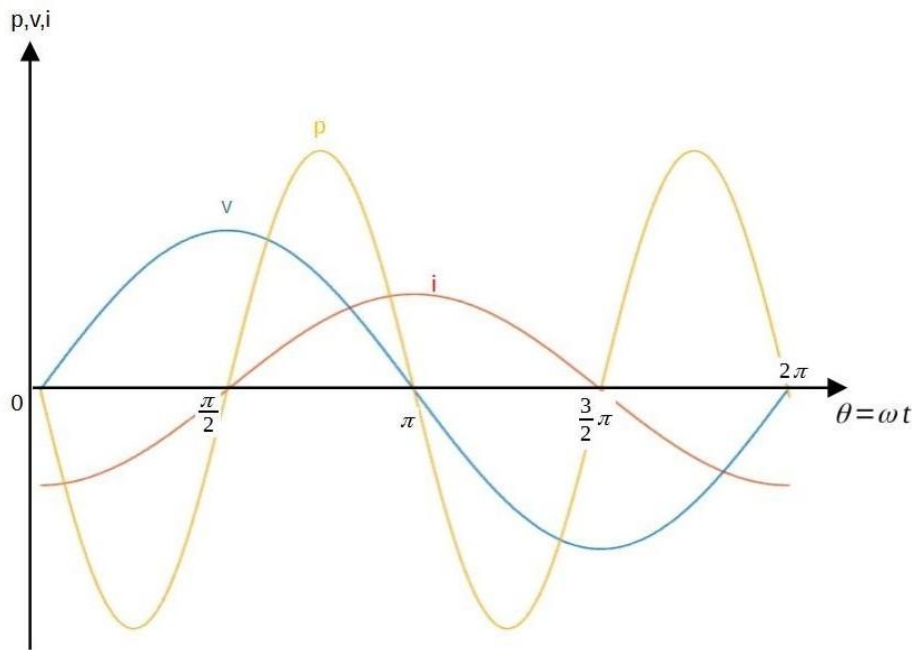


Fig 3.13: Waveforms of v, i and p for Pure inductive circuit

Note: Current lags the voltage by 90° . ϕ is 90° . Hence power factor is zero.

Analysis of a purely capacitive circuit:

Consider a capacitor of pure capacitance C , across which an alternating voltage $v = V_m \sin \omega t$ is applied, due to which an alternating current i flows, charging the plates of the capacitor with a charge of q coulombs as shown in Figure 3.14 (a).

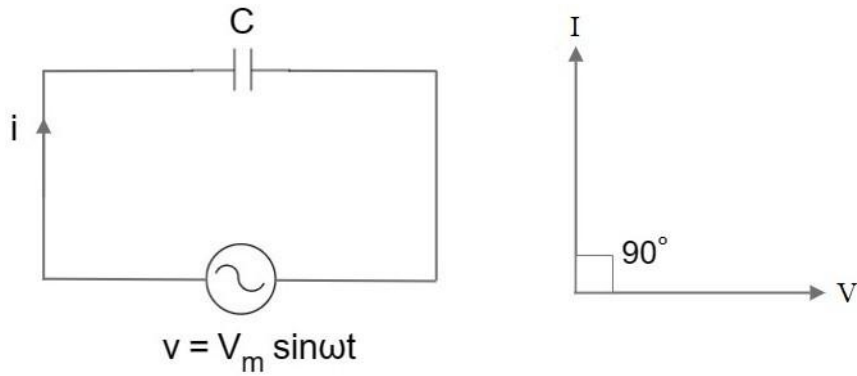


Fig 3.14: a) Pure capacitive circuit b) Phasor Diagram

The current 'i' flowing through the circuit is given by

$$\begin{aligned}
 i &= \frac{dq}{dt} = \frac{d(Cv)}{dt} = C \frac{dv}{dt} = C \frac{d}{dt}(V_m \sin(\omega t)) \\
 &= \omega C V_m \cos \omega t = \frac{V_m}{1/\omega C} \sin\left(\omega t + \frac{\pi}{2}\right) \\
 &= \frac{V_m}{X_c} \sin\left(\omega t + \frac{\pi}{2}\right) \\
 i &= I_m \sin\left(\omega t + \frac{\pi}{2}\right)
 \end{aligned}$$

Where $I_m = \frac{V_m}{X_c}$ and $X_c = \frac{1}{\omega C} = \frac{1}{2\pi f C}$ = Capacitive reactance in Ohms.

By observing the equations for voltage and current, we find that the current leads the voltage by an angle $\pi/2$. Vectorial representation of the rms values of voltage and current is as shown in Figure 3.14 (b).

The instantaneous power is given by, $p = v \cdot i = V_m \sin \omega t \cdot I_m \sin\left(\omega t + \frac{\pi}{2}\right)$

$$p = V_m I_m \sin \omega t \cdot \cos \omega t = \frac{1}{2} V_m I_m \sin 2\omega t$$

The equation for p consists of a quantity which is periodically varying and having a frequency two times the frequency of the applied voltage and whose average value is zero.

The power consumed by a pure capacitor is zero, because the power is a scalar

quantity and only its average value has to be considered. The power given to the pure capacitance is stored in the form of an electrostatic field. The waveforms of v , i and p are as shown in Figure 3.15

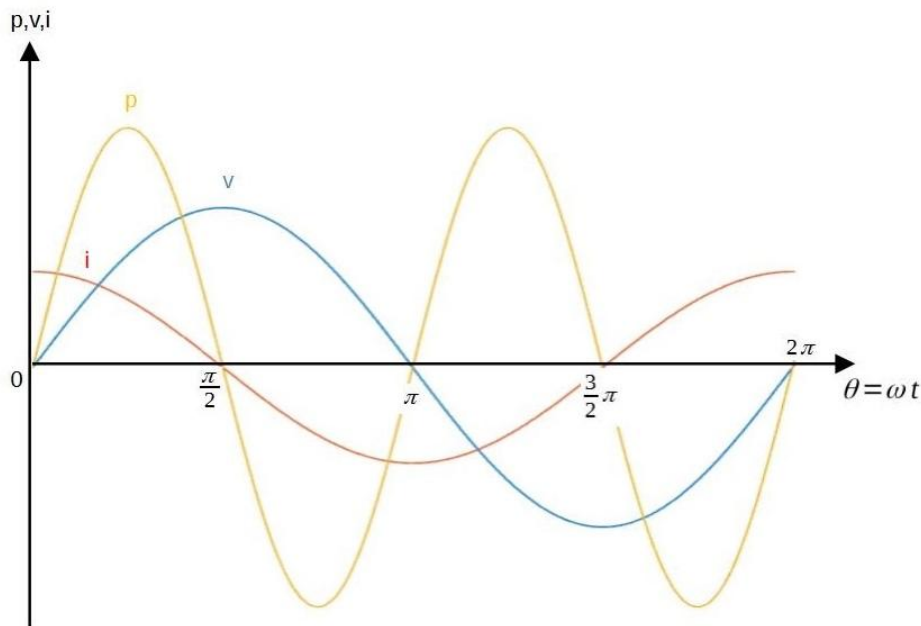


Fig3.15: Waveforms of v , i and p for Pure capacitive circuit

Note: Current leads the voltage by 90° . ϕ is 90° . Hence power factor is zero.

Exercise 1: An AC Circuit consist of a pure resistance of $20\ \Omega$ and is connected across an AC sinusoidal supply of 230 V , 50Hz . Calculate current, power consumed, equation for voltage and current.

$$I = V/R = 230/20 = 11.5\text{A}$$

$$P = VI = 230 \times 11.5 = 2645\text{ W}$$

$$V_m = \sqrt{2}V = 325.27\text{V}$$

$$I_m = \sqrt{2}I = 16.26\text{A}$$

$$\omega = 2\pi f = 314\text{rad/sec}$$

$$\text{Voltage expression is given by, } v = V_m \sin \omega t = 325.27 \sin (314t)\text{ V}$$

$$\text{Current expression is given by, } i = I_m \sin \omega t = 16.26 \sin (314t)\text{ A}$$

Exercise 2: A pure inductive circuit allows a current of 20 A to flow from 230V , 50Hz supply. Calculate inductive reactance, inductance of the coil, power absorbed, equations for voltage and current.

$$\text{Inductive reactance, } X_L = V/I = 230/20 = 11.5\ \Omega$$

$$\text{Coil inductance, } L = X_L / 2\pi f = 11.5 / 2\pi \times 50 = 0.036\text{H}$$

Power absorbed = 0

Peak value of the voltage, $V_m = \sqrt{2}V = \sqrt{2} * 230 = 325.27V$

Peak value of the current, $I_m = \sqrt{2}I = \sqrt{2} * 20 = 28.28A$

$\omega = 2\pi f = 314 \text{ rad/sec}$

Voltage expression is given by, $v = V_m \sin(\omega t) = 325.27 \sin 314t \text{ V}$

Current expression is given by, $i = I_m \sin(\omega t - \frac{\pi}{2}) = 28.28 \sin(314t - \frac{\pi}{2}) A$

Exercise 3: A 400 μ F capacitor is connected across a 230V, 50Hz supply. Calculate capacitive reactance, rms current, equations for the voltage and current.

Capacitive reactance $= X_C = 1/(2\pi fC) = 1/(2\pi * 50 * 400 * 10^{-6}) = 7.95\Omega$

RMS current, $I = V/X_C = 230/7.95 = 28.93A$

Peak value of the voltage, $V_m = \sqrt{2}V = \sqrt{2} * 230 = 325.27V$

Peak value of the current, $I_m = \sqrt{2}I = \sqrt{2} * 28.93 = 40.91A$

$\omega = 2\pi f = 314 \text{ rad/sec}$

Voltage expression is given by, $v = V_m \sin(\omega t) = 325.27 \sin 314t$

Current expression is given by, $i = I_m \sin(\omega t + \frac{\pi}{2}) = 40.91 \sin(314t + \frac{\pi}{2}) A$

Analysis of a R-L Series circuit:

Consider an R-L series circuit as shown in Figure 3.15a. An alternating voltage of $v = V_m \sin \omega t$ is applied, due to which an alternating current i flows through the circuit. The vector diagram considering I as reference vector is as shown in Figure.3.16 b.

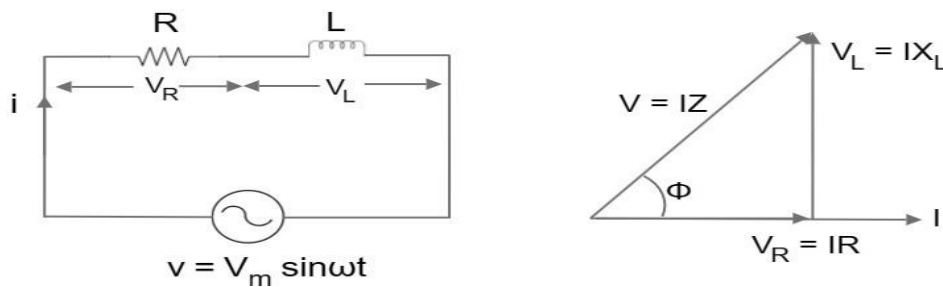


Fig 3.16: a) RL series circuit b) Phasor Diagram

The vector diagram consists of three voltages,

$V_R = I R$ which is in phase with the current,

$V_L = I X_L$ which leads the current by 90.

The vector sum of these two voltages is applied voltage $V = I Z$. here, Z is the impedance of the circuit in ohms. The impedance of an a.c circuit may be defined as the opposition offered for the flow of alternating current in the circuit. It is the phasor sum of resistance and reactance of the circuit. The impedance triangle for an R-L series circuit is shown in Figure 3.17

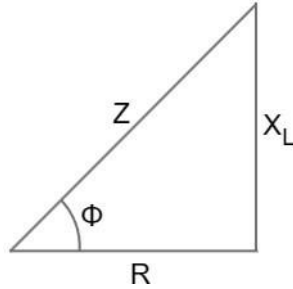


Fig 3.17: Impedance triangle

$$V = V_R + V_L \text{ (phasor sum)}$$

$$V = \sqrt{V_R^2 + V_L^2} = \sqrt{I^2 R^2 + I^2 X_L^2}$$

$$= I \sqrt{R^2 + X_L^2} = IZ$$

$$\text{where, } Z = \sqrt{R^2 + X_L^2} \text{ and } \phi = \tan^{-1}\left(\frac{X_L}{R}\right) = \cos^{-1}\left(\frac{R}{Z}\right)$$

From the phasor diagram shown in Figure.3.16b, we observe that the current lags the voltage by an angle ϕ .

$$\text{If } v = V_m \sin \omega t$$

$$i = I_m (\sin(\omega t) - \phi)$$

The instantaneous power is given by,

$$p = v \cdot i = V_m \sin \omega t I_m (\sin \omega t - \phi)$$

$$= V_m I_m \frac{1}{2} [\cos \phi - \cos(2\omega t - \phi)]$$

$$p = \frac{1}{2} V_m I_m \cos \phi - \frac{1}{2} V_m I_m \cos(2\omega t - \phi)$$

The second term in the equation is a periodically varying quantity, whose frequency is two times the frequency of the applied voltage and its average value is zero. As power is always an average value, only the first term represents the power consumed.

$$\therefore P = \frac{1}{2} V_m I_m \cos \phi = \frac{1}{2} \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} \cos \phi = VI \cos \phi$$

Where, $\cos(\phi)$ is known as the power factor of the circuit. The waveforms of v , i and p are shown in Figure 3.18. The areas of the +ve power lobes is more than the areas of the -ve power lobes, indicating that

the power received by the circuit is more than the power returned by the circuit. Hence, the circuit consumes a net power given by, $P = V I \cos \phi$.

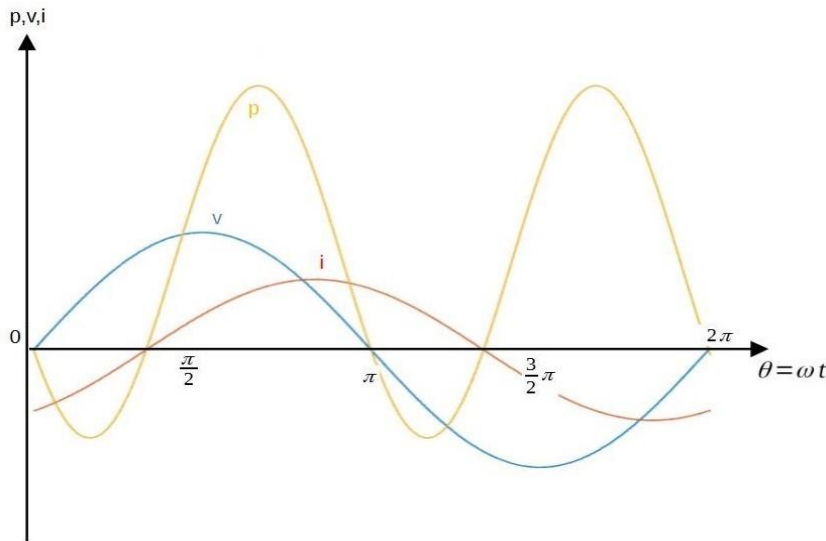


Fig 3.18: Waveforms of v, i and p for RL series circuit

Analysis of a R-C Series circuit

Consider an R-C series circuit as shown in Figure.3.19a. to which an alternating voltage of $v = V_m \sin(\omega t)$ is applied, due to which an alternating current i flows through the circuit. I is taken as the reference vector. The diagram consists of three voltages,

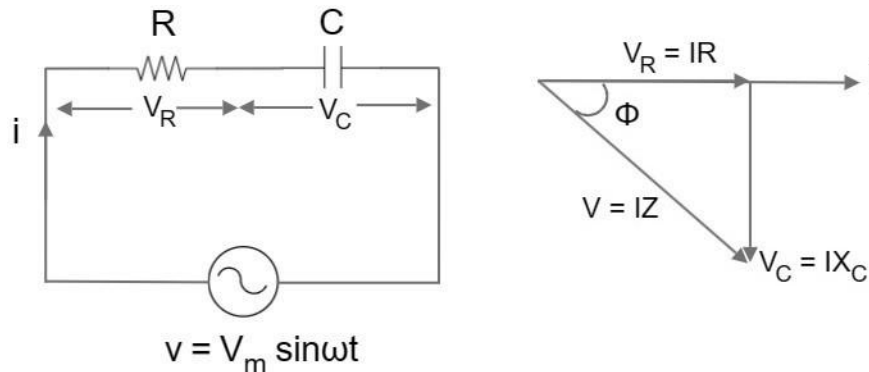


Fig 3.19: a) RC series circuit b) Phasor Diagram

$V_R = I R$ which is in phase with the current, $V_C = I X_C$ which lags the current by 90° . The vector sum of these two voltages is applied voltage $V = IZ$. Here, Z is the impedance of the circuit in ohms. The impedance triangle for an R-C series circuit is shown in Figure 3.20.

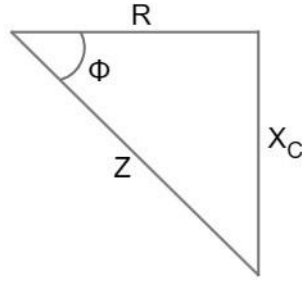


Fig 3.20: Impedance Diagram

$V = V_R + V_C$ (phasor sum)

$$V = \sqrt{V_R^2 + V_C^2} = \sqrt{I^2 R^2 + I^2 X_C^2}$$

$$= I \sqrt{R^2 + X_C^2} = IZ$$

Where, $Z = \sqrt{R^2 + X_C^2}$ and $\phi = \tan^{-1}\left(\frac{X_C}{R}\right) = \cos^{-1}\left(\frac{R}{Z}\right)$

The current leads the voltage by an angle ϕ

If $v = V_m \sin \omega t$, Then, $i = I_m \sin(\omega t + \phi)$

The current in the circuit is given by, $I = V/Z$

The impedance triangle for an R-C series circuit is shown in Figure 3.20. The instantaneous power is given by,

$$p = v \cdot i = V_m \sin(\omega t) \cdot I_m \sin(\omega t + \phi)$$

$$= V_m I_m \frac{1}{2} [\cos(-\phi) - \cos(2\omega t + \phi)]$$

$$p = \frac{1}{2} V_m I_m \cos(\phi) - \frac{1}{2} V_m I_m \cos(2\omega t + \phi)$$

The second term in the equation is a periodically varying quantity, whose frequency is two times the frequency of the applied voltage and its average value is zero. As power is always an average value, only the first term represents the power consumed.

$$\therefore P = \frac{1}{2} V_m I_m \cos(\phi) = \frac{1}{2} \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} \cos(\phi) = VI \cos(\phi)$$

The waveforms of v , i and p are shown in Figure.3.21.

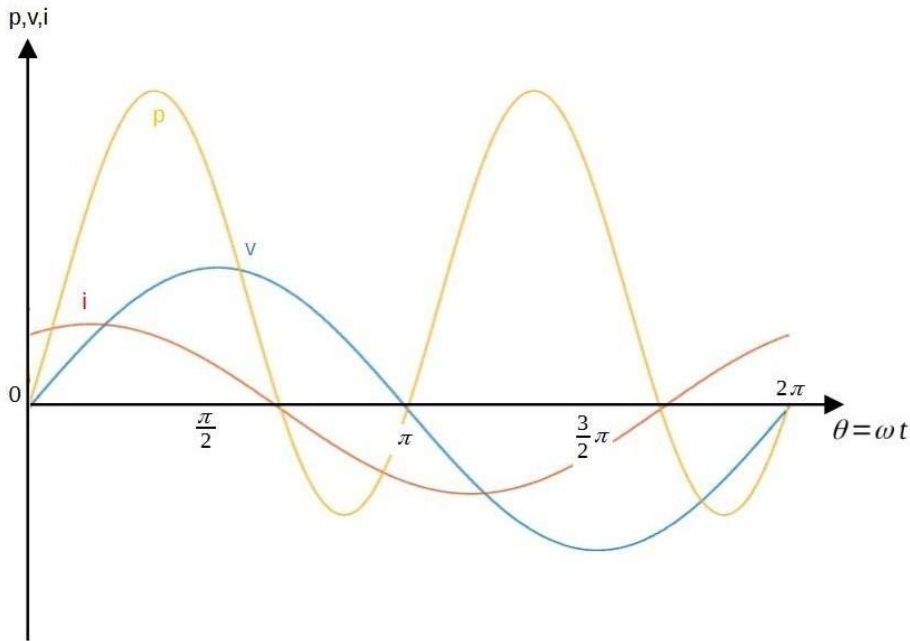


Fig 3.21: Waveforms of v, i and p for RC series circuit

Analysis of a R-L-C Series circuit

Consider an R-L-C series circuit as shown in Figure.3.22. to which an alternating voltage of $v = V_m \sin \omega t$ is applied, due to which an alternating current i flows through the circuit. Three cases of the circuit can be discussed here.

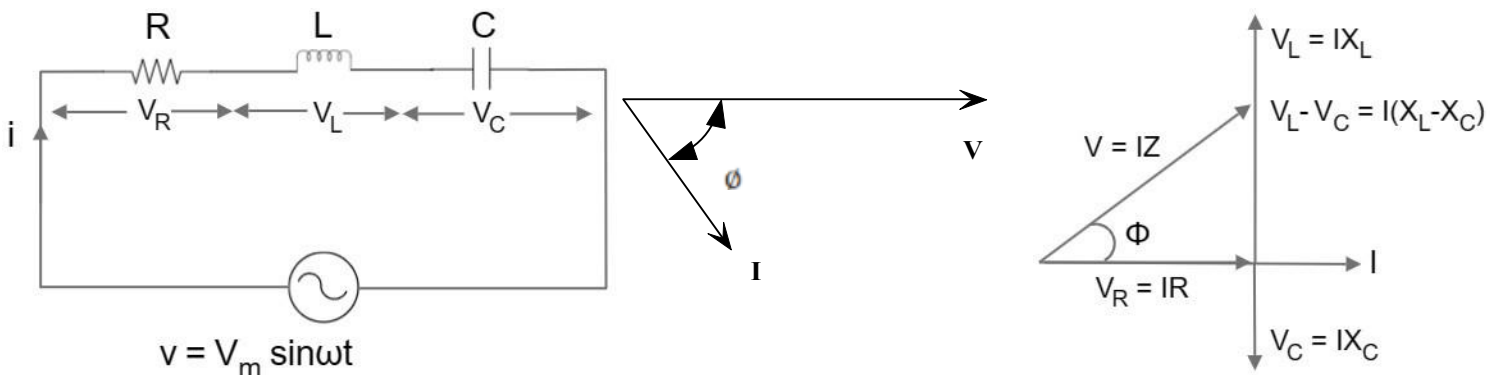


Fig 3.22: a) RLC series circuit b) Phasor Diagram for $X_L > X_C$ c) Impedance triangle

Case 1: When $X_L > X_C$

When the inductive reactance is more than the capacitive reactance, the vector diagram of the circuit as shown in Figure.3.20b. From the vector diagram, we observe that the current lags the voltage by an angle ϕ . The impedance triangle is shown in Figure.3.22c

$$I = \frac{V}{Z}$$

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

The circuit behaves similar to an R-L circuit. If $v = V_m \sin(\omega t)$, then $i = I_m \sin(\omega t - \phi)$

Hence it can be proved that the power consumed is given by $P = VI \cos \phi$.

Case 2: When $X_L < X_C$

When the inductive reactance is less than the capacitive reactance, the vector diagram of the circuit as shown in Figure 3.23a. From the vector diagram, we observe that the current leads the voltage by an angle ϕ .

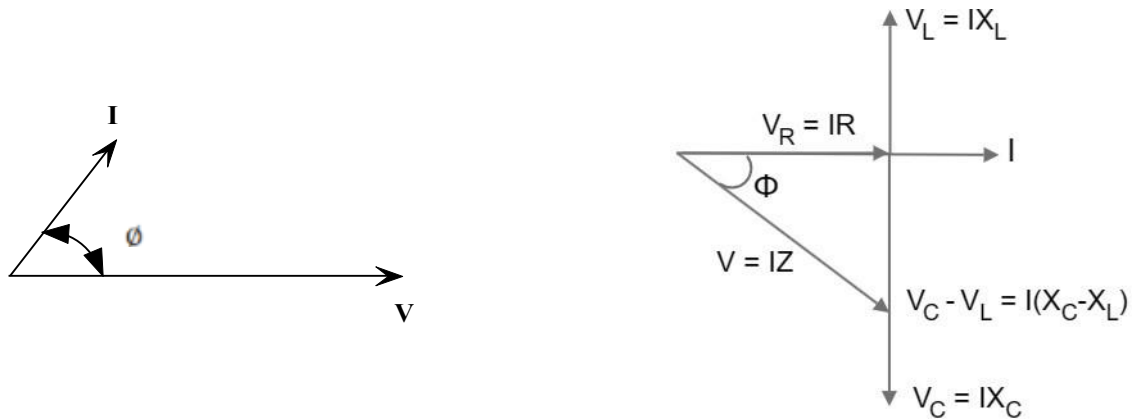


Fig 3.23 a) Phasor Diagram for $X_L < X_C$ b) Impedance triangle

$$I = \frac{V}{Z}$$

$$Z = \sqrt{R^2 + (X_C - X_L)^2}$$

The circuit is similar to an R-C circuit.

If $v = V_m \sin(\omega t)$, then $i = I_m \sin(\omega t + \phi)$

Hence it can be proved that the power consumed is given by $P = VI \cos \phi$.

Case 3: When $X_L = X_C$

When the inductive reactance is equal to the capacitive reactance, the vector diagram of the circuit as shown in Figure. 3.24.

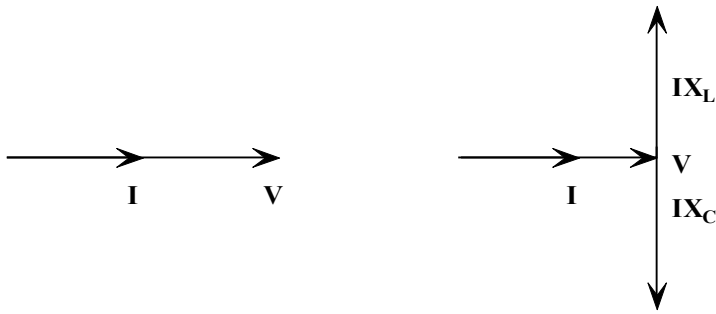


Fig 3.24 a) Phasor Diagram for $X_L = X_C$ b) Impedance triangle

V_L and V_C cancel out each other. The current is in phase with the voltage and circuit behaves as a pure resistive circuit. Hence $Z = R$. The current is in phase with voltage i.e: $\phi = 0$ (Unity power factor)

If $v = V_m \sin(\omega t)$, then, $i = I_m \sin(\omega t)$

Hence it can be proved that the power consumed is given by $P = VI$, which is proved in the case of a pure resistance circuit.

Exercise 4: A resistance of 10Ω is connected in series with a pure inductance of 55mH and the circuit is connected to a 230V , 50Hz sinusoidal supply. Calculate the a) circuit current b) Phase angle

Given data: $R = 10 \Omega$ $L = 55\text{mH}$ $V = 230\text{V}$, $f = 50\text{Hz}$

a) Current $I = V/Z$

$$X_L = 2\pi fL = 2 \times \pi \times 50 \times 55 \times 10^{-3} = 17.27 \Omega$$

$$\text{Impedance } Z = \sqrt{R^2 + X_L^2}$$

$$\text{Substituting for } X_L, Z = \sqrt{10^2 + 17.27^2} = 19.95 \Omega.$$

$$\therefore \text{Current } I = 230/19.95 = 11.52\text{A}$$

$$\text{b) Phase angle } \phi = \tan^{-1} \left(\frac{X_L}{R} \right) = \tan^{-1} \left(\frac{17.27}{10} \right) = 60^\circ \text{ lag}$$

Exercise 5: A capacitor of $16\mu\text{F}$ takes a current of 2A when the alternating voltage applied across it is 230V . Calculate: (a) the frequency of the applied voltage (b) the resistance to be connected in series with the capacitor to reduce the current in the circuit to 1A at the same frequency (c) the phase angle of the resulting circuit.

$$\text{a) } X_C = \frac{V}{I} = \frac{230}{2} = 115\Omega$$

$$X_C = \frac{1}{2\pi fC}$$

$$\therefore f = \frac{1}{2\pi CX_C} = \frac{1}{2\pi \times 16 \times 10^{-6} \times 115} = 86.5 \text{ Hz.}$$

b) when a resistance is connected in series with the capacitor,

$$c) Z = \frac{V}{I} = \frac{230}{1} = 230 \Omega$$

$$Z = \sqrt{R^2 + X_C^2} = \sqrt{R^2 + 115^2} = 230 \Omega$$

$$\text{hence } R = 199.18 \Omega$$

$$c) \phi = \cos^{-1} \frac{R}{Z} = \cos^{-1} \frac{199.18}{230} = 30^\circ \text{ lead (RC circuit)}$$

Exercise 6: A 230V, 50Hz AC supply is applied to a coil of 0.12 H inductance and 5 Ω resistance, the connection in series with a 12 μ F capacitance. Calculate the impedance, current, phase angle between voltage and current, power factor and the power consumed.

$$\text{Inductive reactance, } X_L = 2\pi fL = 2\pi \times 50 \times 0.12 = 37.69 \Omega$$

$$\text{Capacitive reactance, } X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi \times 50 \times 12 \times 10^{-6}} = 265.25 \Omega$$

$$Z = \sqrt{R^2 + (X_C - X_L)^2} = 227.6 \Omega$$

$$I = V/Z = 230/227.6 = 1.01 \text{ A}$$

To find the phase angle of the resulting circuit:

$$\tan \phi = (X_C - X_L)/R = 227.56/5 = 45.51$$

$$\text{Therefore } \phi = 88.74^\circ \text{ lead}$$

$$\cos \phi = R/Z = 5 / 227.6 = 0.0219 \text{ lead}$$

$$P = VI \cos \phi = 230 \times 1.01 \times 0.0219 = 5.08 \text{ W}$$

Exercise 7: A circuit having resistance of 6 Ω an inductance of 0.075 Henry and a capacitance of 50 μ F in series is connected across a 100V, 50Hz supply. Calculate a) the impedance b) the current c) the voltage across R,L and C d) the phase difference between the current and the supply voltage.

$$\text{Data given: } R=6 \Omega, L=0.075\text{H}, C=50\mu\text{F}, V=100\text{V}, f=50\text{Hz}$$

$$X_L = 2\pi fL = 23.56 \Omega$$

$$X_C = 1/2\pi fC = 63.66 \Omega$$

$$Z = \sqrt{R^2 + (X_C - X_L)^2} = 40.54 \Omega$$

$$\text{Current } I = V/Z = 100 / 40.54 = 2.466 \text{ A}$$

$$\text{Voltage across resistance } V_R = IR = 2.466 \times 6 = 14.76 \text{ V}$$

$$\text{Voltage across inductance } V_L = IX_L = 2.466 \times 23.56 = 58.09 \text{ V}$$

$$\text{Voltage across capacitance } V_C = IX_C = 2.466 \times 63.66 = 156.98 \text{ V}$$

$$\text{Phase difference } \phi = \cos^{-1} \left(\frac{R}{Z} \right) = \cos^{-1} \left(\frac{6}{40.54} \right) = 81.48^\circ \text{ lead}$$